IEOR 221 - HW 1 - Jie Wen

1) Monthly Payment for a Car Loan

- **Principal (P):** $20,000
- **Annual Interest Rate (r):** $10\% = 0.10$
- **Loan Duration (T):** 5 years
- **Number of Monthly Payments (n):** $5 \times 12 = 60$
- **Monthly Interest Rate ($r_{monthly}$):** $r/12 = 0.10/12$

The formula for the present value of an annuity (loan principal) is:
$P = M \sum_{i=1}^{n} \frac{1}{(1+r_{monthly})^i} = M \frac{1-(1+r_{monthly})^{-n}}{r_{monthly}}$

Rearranging to solve for the monthly payment (M):
$M = P \frac{r_{monthly}}{1-(1+r_{monthly})^{-n}} = 20000 \times \frac{0.10/12}{1-(1+0.10/12)^{-60}} \approx 424.94$

The monthly payment is **$424.94**.

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2) 2-Year Par Yield

Given zero rates with semiannual compounding:

- 6-month: 5%
- 12-month: 6%
- 18-month: 6.5%
- 24-month: 7%

The par yield c (with semiannual coupons) satisfies:

$$100 = 50c \cdot (DF(0.5) + DF(1.0) + DF(1.5) + DF(2.0)) + 100 \cdot DF(2.0)$$

where discount factors are:

$$DF(0.5) = \frac{1}{(1 + 0.05/2)^1} = \frac{1}{1.025} \approx 0.97560976$$

$$DF(1.0) = \frac{1}{(1 + 0.06/2)^2} = \frac{1}{1.03^2} \approx 0.942595901$$

$$DF(1.5) = \frac{1}{(1 + 0.065/2)^3} = \frac{1}{1.0325^3} \approx 0.90851022$$

$$DF(2.0) = \frac{1}{(1 + 0.07/2)^4} = \frac{1}{1.035^4} \approx 0.87144222$$

Sum of discount factors:

$$\sum DF \approx 0.97560976 + 0.942595901 + 0.90851022 + 0.87144222 = 3.698158101$$

$$c = \frac{2 \cdot (1 - DF(2.0))}{\sum DF} = \frac{2 \cdot (1 - 0.87144222)}{3.698158101} \approx 0.069525$$

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3) Forward Interest Rates

- **Forward rate for the second year (period $[1, 2]$ years):**
  $T_1 = 1, r_1 = 0.02$
  $T_2 = 2, r_2 = 0.03$
  $r_{1,2} = \frac{(0.03 \times 2) - (0.02 \times 1)}{2-1} = 0.04$
  Forward rate for the second year = **4.00%**
- **Forward rate for the third year (period $[2, 3]$ years):**
  $T_1 = 2, r_1 = 0.03$
  $T_2 = 3, r_2 = 0.037$
  $r_{2,3} = \frac{(0.037 \times 3) - (0.03 \times 2)}{3-2} = 0.051$
  Forward rate for the third year = **5.10%**
- **Forward rate for the fourth year (period $[3, 4]$ years):**
  $T_1 = 3, r_1 = 0.037$
  $T_2 = 4, r_2 = 0.042$
  $r_{3,4} = \frac{(0.042 \times 4) - (0.037 \times 3)}{4-3} 0.057$
  Forward rate for the fourth year = **5.70%**

- **Forward rate for the fifth year (period $[4,5]$ years):**
  $T_1 = 4, r_1 = 0.042$
  $T_2 = 5, r_2 = 0.045$
  $r_{4,5} = \frac{(0.045\times5)-(0.042\times4)}{5-4} = 0.057$
  Forward rate for the fifth year = **5.70%**

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4) Treasury Zero Rates

Given cash prices:

- 6-month T-bill: 94.0
- 1-year T-bill: 89.0
- 1.5-year bond: 94.84 (coupon \$4 every 6 months)
- 2-year bond: 97.12 (coupon \$5 every 6 months)

Assume continuous compounding.

- **6-month rate $r_{0.5}$:**

$$94 = 100e^{-r_{0.5}\cdot0.5} \implies r_{0.5} = -2\ln(0.94) \approx 12.375\%$$

- **1-year rate r_1:**

$$89 = 100e^{-r_1\cdot1} \implies r_1 = -\ln(0.89) \approx 11.653\%$$

- **1.5-year rate $r_{1.5}$:**

$$94.84 = 4e^{-r_{0.5}\cdot0.5} + 4e^{-r_1\cdot1} + 104e^{-r_{1.5}\cdot1.5} \implies e^{-1.5r_{1.5}} = \frac{87.52}{104} \approx 0.841538 \implies r_{1.5} \approx \frac{0.17254}{1.5} \approx 11.502\%$$

- **2-year rate r_2:**

$$97.12 = 5e^{-0.5r_{0.5}} + 5e^{-r_1} + 5e^{-1.5r_{1.5}} + 105e^{-2r_2} \implies e^{-2r_2} = \frac{83.76231}{105} \implies r_2 \approx 11.299\%$$

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5) Equivalent Rates

Given: 5% per annum with semiannual compounding.

Effective semiannual rate: 2.5%. Effective annual rate:

$$(1+0.025)^2 - 1 = 5.0625\%$$

- **(a) Annual compounding:**
  Equivalent to effective annual rate: 5.0625%
- **(b) Monthly compounding:**
  Let r_m be the monthly compounded rate:

$$(1+\frac{r_m}{12})^{12} = 1.050625 \implies r_m \approx 4.949\%$$

- **(c) Continuous compounding:**
  Let r_c be the continuously compounded rate:

$$e^{r_c} = 1.050625 \implies r_c = \ln(1.050625) \approx 0.049385 = 4.9385\%$$